RSA Security and Common Factors

25. Write a C program for set of blocks encoded with the RSA algorithm and we don't have the private key. Assume n = pq, e is the public key. Suppose also someone tells us they know one of the plaintext blocks has a common factor with n. Does this help us in any way?

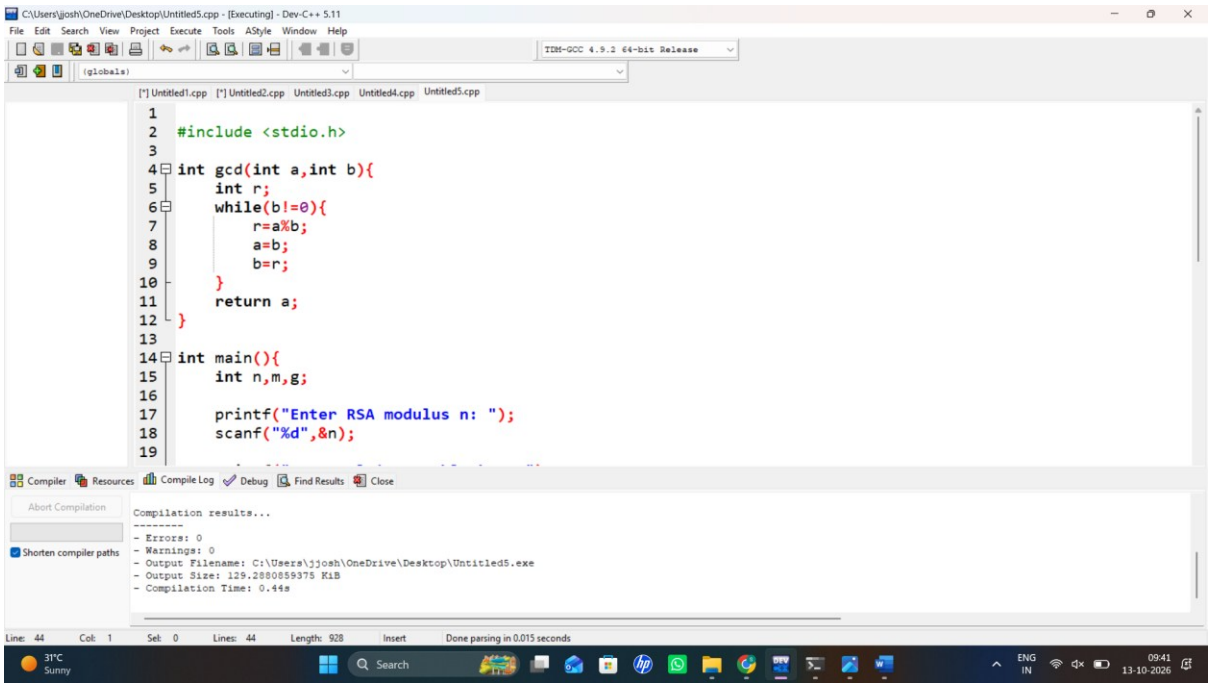
Aim

To write a C program to determine whether a plaintext block shares a common factor with the RSA modulus \(n\), and to understand how this information can affect RSA security when the private key is unknown.

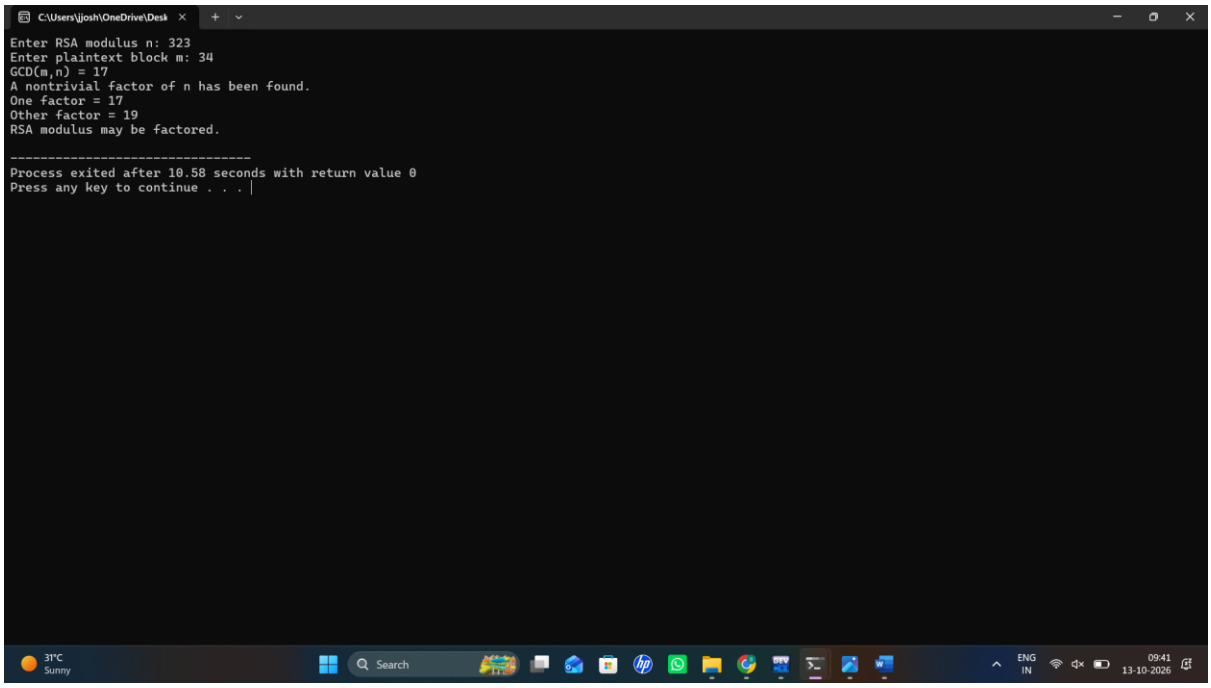
Algorithm

1. Start the program.
2. Read the RSA modulus \(n\) and a plaintext block \(m\).
3. Calculate the greatest common divisor (GCD) of \(m\) and \(n\) using the Euclidean algorithm.
4. If \(\gcd(m,n)=1\), the block does not reveal a nontrivial common factor.
5. If \(1<\gcd(m,n)<n\), the GCD reveals a nontrivial factor of \(n\).
6. Display the result and explain its significance for RSA security.
7. Stop the program.

Program :



Output:



Result

Thus, the C program successfully determines whether a plaintext block shares a nontrivial factor with the RSA modulus.

Answer to the theoretical question:

Yes, this information can help. If \(n=pq\) and the plaintext block \(m\) shares a factor with \(n\), calculating \(\gcd(m,n)\) may reveal \(p\) or \(q\). Once the modulus is factored, an attacker can calculate \(\phi(n)\) and determine the private exponent \(d\), compromising the RSA key.

For example,

\[n=323=17\times19 \]

\[\gcd(34,323)=17 \]

Knowing this factor allows the other factor, \(19\), to be calculated. However, if the GCD equals \(1\), this particular method does not reveal a nontrivial factor.